MIL-L-52046A(ME)
1 July 1981
SUPERSEDING
MIL-L-52046(CE)
2 July 1958

MILITARY SPECIFICATION

LEVEL, CIRCULAR, 2.5 CM DIAMETER,

MOUNTED IN CYLINDER

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cylinder-mounted circular level.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5210

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MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications, and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B209 - Aluminum-Alloy Sheet and Plate.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The circular level shall be constructed as shown on figure 1 and as specified herein. The circular level vial shall be securely attached to the cylinder by any convenient means, provided the vial cannot be displaced by normal use or dropping, storage, or packaging. The vial shall have a nominal diameter of 2.5 cm, a sensitivity of approximately 20 minutes of arc per 2 mm

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displacement and the reference circle shall have a diameter of approximately 8 mm. The glass in the vial shall possess sufficient strength to withstand the stresses produced by sudden temperature changes. Each level furnished shall be in adjustment so that when the bubble is centered, the base of the cylinder is horizontal and level with the bubble. A circle shall be inscribed on the level for centering the bubble, and the bubble shall be centered in this circle when the level is placed on a flat level surface.

3.2 First article (preproduction model). The contractor shall furnish one circular level as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce circular levels that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3.1 Aluminum-alloy sheet. Aluminum-alloy sheet shall conform to ASTM B209, alloy 6061, temper T6.

3.4 Environmental conditions. The circular level shall be capable of satisfactory performance in any ambient temperature from plus 125° F to minus 65° F. There shall be no leakage or solidification of the liquid in the level vial or any other deleterious effects upon the circular level when tested in accordance with 4.2.1.

3.5 Identification marking. Each level shall be identified in accordance with MIL-STD-130.

3.6 Treatment and painting. Exterior metal surfaces shall be treated and painted before vial is secured, in accordance with MIL-T-704, type A.

3.7 Workmanship. Tolerances and gages for metal fits shall conform to the limits specified herein and otherwise to the standards of good commercial practice. Finished contact and bearing surfaces shall be true and exact.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.

4.2.1 Examination. The preproduction circular level shall be examined as specified in 4.3.2. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests. The preproduction circular level shall be tested as specified in 4.4.1 and 4.4.2. Failure of either test shall be cause for rejection.

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the following defects:

Major

101. Nonconformance to any dimensions shown on figure 1.
102. Improper and loose fitting of vial or retaining plate.
103. Treatment and painting not as specified.
104. Poor appearance of welds (porosity, flux inclusions, excessive spatter, undercutting, lack of fusion with parent metal or between parts).
105. Sharp, ragged, or rough edges.

Minor

201. Identification marking not as specified.
202. Workmanship not as specified.

4.4 Tests.

4.4.1 Temperature test. The preproduction model shall be placed in a temperature test chamber and held at a temperature of not less than 125° F for a period of 3 hours. The preproduction model shall then be placed in a test chamber and held at a temperature of minus 65° F for a period of 3 hours. Nonconformance to 3.4 shall constitute failure of this test.

4.4.2 Leveling test. The circular level shall be placed on a flat level surface. The bubble and the circle inscribed on the top of the level shall be in alinement. Nonconformance to 3.1 shall constitute failure of this test.

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4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

Major

106. Materials, methods, or containers not as specified for level A or B. Each incorrect material, method, or container shall constitute one defect.
107. Consolidation not as specified for level A.
108. Quantities packed together exceed that specified for level A or B.
109. Box closure not as specified for level B.
110. Preservation and packing not in accordance with the referenced document as specified for commercial.
111. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each level shall be preserved in accordance with MIL-P-116, Method IC-2 using the contractor's standard commercial box or a close fitting box conforming to PPP-B-566.

5.1.1.1 Consolidation. Levels shall be consolidated together in quantities not to exceed the weight limitation of the box, in close fitting boxes conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. Levels shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Levels preserved as specified in 5.1, shall be packed together in close fitting boxes conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 200 pounds. Box closure and strapping shall be in accordance with the appendix to the box specification.

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5.2.2 Level B. Levels preserved as specified in 5.1, shall be packed together in quantities not to exceed the size or weight limitation of the box, in close fitting boxes conforming to PPP-B-636, V3c, style optional. Box closure and sealing shall be as specified for method V in the appendix to the box specification.

5.2.3 Commercial. Levels preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for Military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Comnercial. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The level is intended for use in leveling portable airport light hoods.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model, and number of circular levels required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification. (See 3.3).

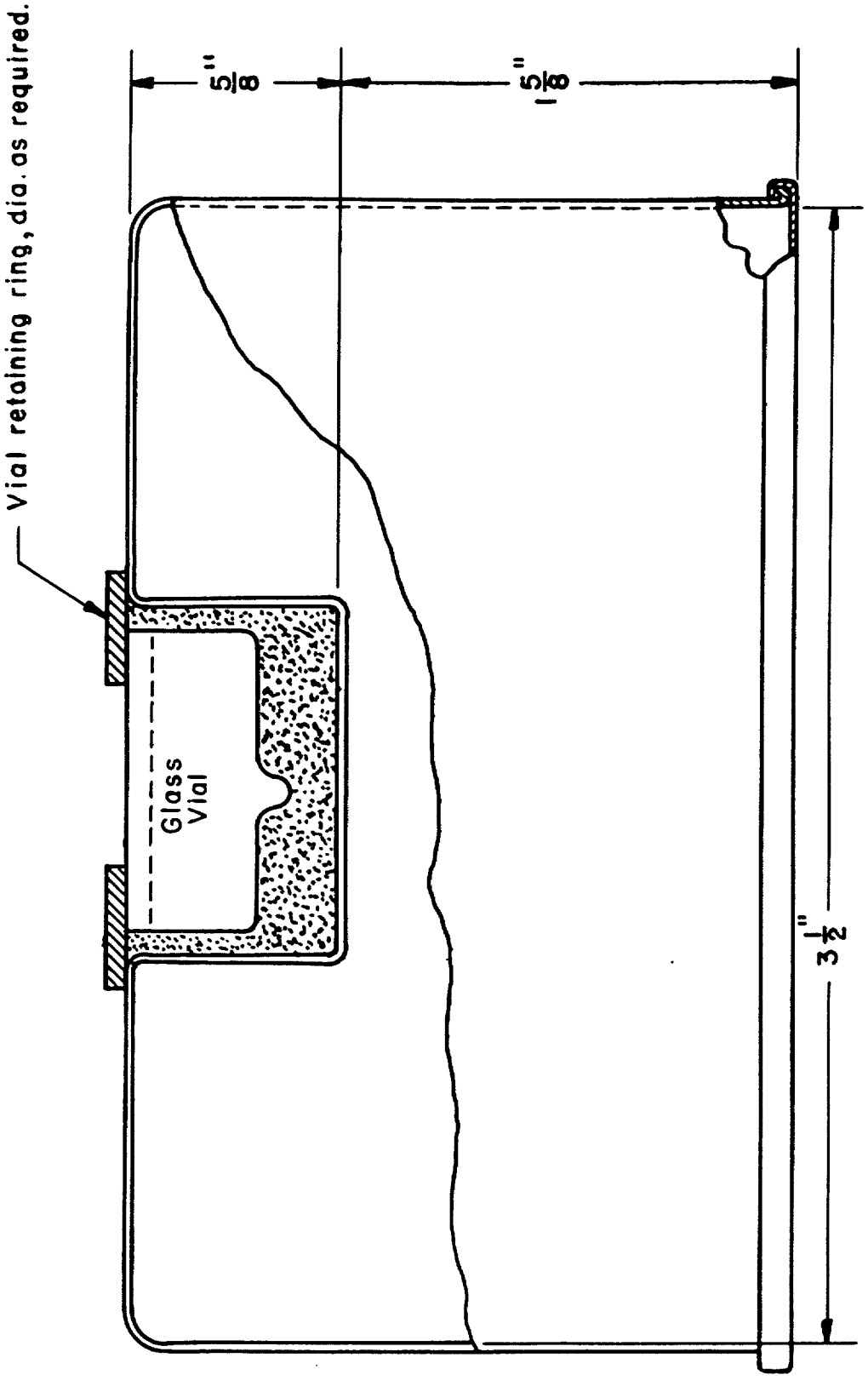
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Figure 1. Leveling Device
Aluminum Alloy, 24 Ga.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-L-52046A(ME) Level, Circular, 2.5 CM Diameter, Mounted in Cylinder

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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